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# Cadmium and mercury-induced hyperglycemia in the fresh water crab, *Oziotelphusa senex senex*: Involvement of neuroendocrine system

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ABSTRACT

The effect of exposure to sublethal concentrations of cadmium chloride and mercuric chloride on hemolymph glucose levels of the freshwater crab, *Oziotelphusa senex senex*, was studied. Intact crabs exposed to cadmium or mercury exhibited a significant hyperglycemia compared to controls, but no significant differences in hemolymph glucose level were detected among the eyestalkless crabs after exposure to metals, suggesting that the effect of metals could be on the sinus gland in the eyestalks, increasing secretion of the hyperglycemic hormone. To test this hypothesis, eyestalks were collected from control and metal exposed crabs, and tested for hyperglycemic effect and also for the hyperglycemic hormone levels. The levels of hyperglycemic hormone and the hyperglycemic effect were significantly low in the eyestalks collected from metal exposed crabs when compared with eyestalks from control crabs. These results strongly suggest that metals act, at least in part, by triggering the secretion of hyperglycemic hormone from the eyestalk.



## 1. Introduction

Ever since the outbreak of 'Minimata' and 'Itai-Itai' diseases due to consumption of fish and other seafood contaminated with mercury and cadmium, respectively (Ui, 1972), an increasing awareness of their toxicity has prompted the monitoring of mercury and cadmium contamination and of their detrimental effects on aquatic organisms. Metals because of their bioaccumulation, immutable and non-degradable properties, if exposure persists over time, even at low concentrations, are reported to cause severe harmful effects in aquatic organisms. The freshwater rice field crab, *Oziotelphusa senex senex* has been extensively used as a bio-indicator of the effects of pollutants (Reddy et al., 1982a,b, 1983; Bhagyalakshmi et al., 1983). The earlier research studies on crabs also include the effect of metals on hemolymph sugar level (Reddy and Bhagyalakshmi, 1994) and enzyme activities (Reddy, 1997, 1999).

In crustaceans, hemolymph sugar level is under the regulation of an eyestalk factor, called the crustacean hyperglycemic hormone (CHH). CHH, which was first named as a diabetogenic factor by Abramowitz et al. (1944), is a neuropeptide produced in the X-organ-sinus gland complex. Immunocytochemical studies, using antisera against CHHs from several species, have consistently mapped the CHH to a cluster of large neurosecretory perikarya in the X-organ (Gorgels-Kallen et al., 1982; Leuven et al., 1982; Martin et al., 1984; Van Herp et al., 1984; Rotllant et al., 1993). The mode and site of action of CHH has been thoroughly established (Sedlmeier, 1985; Keller et al., 1985). Hohnke and Scheer (1970) suggested that the primary role of hyperglycemic hormone is not to elevate hemolymph sugar level, but to elevate intracellular glucose through the degradation of glycogen by activating phosphorylase. These glucose molecules leak into hemolymph thereby resulting in hyperglycemia. CHH amino acid sequences from a number of crustaceans have been determined by Edman degradation and in some cases, also by cloning of precursor cDNAs (for reviews see Keller, 1992; Chang, 1993; De Kleinjn and Van Herp, 1995). Recently several isoforms of CHH were discovered in several crustaceans including *O. senex senex* (Reddy and Reddy, 2006a,b).

In contrast to considerable accumulated knowledge on the structure of CHH, its localization in neurosecretory cells and its site and mode of action, relatively little is known on the regulation of synthesis and release of CHH. Hyperglycemia is commonly observed under a variety of stressful conditions and there is a considerable indirect evidence to suggest that this stress hyperglycemia is caused by the release of CHH (Keller and Sedlmeier, 1988). The role of CHH in stress-induced blood sugar changes has only recently begun. Reddy et al. (1994) provided evidence that CHH mediates cadmium-induced hyperglycemia in the red swamp crayfish, *Procambarus clarkii*. Consequently, the present study was designed to assess (1) the effect of cadmium and mercury on

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hemolymph sugar level in the freshwater edible crab *O. senex senex*, (2) whether eyestalk hormone(s) has a role, if metals do induce hyperglycemia and (3) the effects, if any, of cadmium and mercury on CHH levels in the eyestalks of the crab. Further, the study also describes the uptake of cadmium and mercury in different tissues of crab 24 h after exposure to metals.

## 2. Materials and Methods

### 2.1. Collection of animals and maintenance

Adult male crabs, *O. senex senex*, with a body weight of $30 \pm 3$ g and body size of $39 \pm 3$ mm were collected from irrigation canals around Tirupati (13°36′N, 79°21′E), Andhra Pradesh, India. Animals were housed 4 per glass aquaria (length:width: height = 60:30:30 cm) with 40 L sand-filtered tap water. They were acclimatized to laboratory conditions (Temp. 23–25 °C and photo-period 12 L:12 D) for a week before use. The crabs were fed daily with sheep meat and feeding was stopped 24 h before commencement of the experiment.

### 2.2. Test chemicals

Cadmium chloride and mercuric chloride ( > 99.5% purity) were purchased from the Sigma Chemical Co. (St. Louis, MO, USA). A stock solution of 10 g/L was prepared weekly in distilled water, from which small aliquots were taken to achieve the desired concentrations in the exposure tanks. The concentration of metals in exposure tanks were determined before and 24 h after introducing the animals using Atomic Absorption Spectrophotometer (Shimadzu AA 6300).

Experiment I: *Effects of cadmium and mercury on hemolymph glucose level and uptake of metals in tissues of intact crabs*.

The crabs were divided into three groups each consisting of 10 crabs: crabs in the first group were used as controls. Crabs in second and third groups were exposed to 1/5th $LC_{50}$/48 h concentration of cadmium chloride (nominal concentration, 0.4 mg/L; actual concentration, 0.24 ppm cadmium) and mercuric chloride (nominal concentration, 0.1 mg/L; actual concentration, 0.07 ppm mercury), respectively (Kumar Babu et al., 2009). Twenty four hours after exposure, hemolymph was collected and used for glucose determination. The hepatopancreas, chelate leg muscle and gill tissues of crabs from second and third groups were isolated and used for determination of cadmium and mercury contents.

Experiment II: *Effects of cadmium and mercury on hemolymph glucose level in eyestalk ablated crabs*.

To investigate the involvement of eyestalk hyperglycemic hormone in metal-induced hyperglycemia, eyestalks from 45 crabs were ablated. Bilateral eyestalk ablation was achieved by cutting the eyestalk at their bases at the arthrodial membrane. We routinely achieve more than 95% survival of the crabs following the operation. Eyestalkless crabs (ESX) were then evenly divided into three groups. The crabs in Group I, the ESX group, were maintained in tap water, the crabs in the Group II were exposed to 1/5th $LC_{50}$/48 h concentration (0.24 mg/L) of cadmium and the crabs in the III Group were exposed to 1/5th $LC_{50}$ /48 h concentration (0.07 mg/L) of mercury. Hemolymph was collected after 24 h from all the crabs and used for the measurement of glucose.

Experiment III: *Effect of cadmium and mercury on hyperglycemic action of eyestalk extracts*.

To investigate the involvement of the eyestalk hyperglycemic hormone in metal-induced hyperglycemia, eyestalks were collected from control and metal-exposed crabs after 24 h exposure. Eyestalk extracts were prepared by homogenizing the eyestalk neural tissue in crab physiological saline (0.2 M NaCl, 5.4 mM KCl, 2.6 mM $MgCl_2$, 13.5 mM $CaCl_2$, 5.6 mM maleate, 10.8 mM Tris; pH 7.4) and then centrifuging at 10,000*g* for 10 min at 4 °C. The supernatant was used for injections. Uninjured intact crabs were injected with 2.0 eyestalk equivalents in a 25 μL volume. Crabs in the control group received 25 μL of crustacean saline by injection. Hemolymph was drawn from these crabs 2 h after injection and glucose levels were determined.

Experiment IV: *Effect of cadmium and mercury on the levels of hyperglycemic hormone in the eyestalks*.

Eyestalks were collected from control and metal-exposed crabs (24 h post-exposure) and sinus glands were isolated. Following HPLC of sinus gland sample (Reddy and Reddy, 2006a,b), collected fractions were assayed for CHH. CHH levels were measured using ELISA developed by Chang et al. (1998). The polyclonal antibodies for *O. senex senex* CHH were raised in our laboratory. Hyperglycemic hormone was purified from 200 crab sinus glands by means of HPLC (Reddy and Reddy, 2006a,b). It was injected to a rabbit with Freund's complete adjuvant. A booster injection was administered 4 weeks later. After another 2 weeks, the serum was obtained and frozen at −70 °C. The fractions containing IgGs were separated according to the method described earlier (Chang et al., 1998) and used for the ELISA.

### 2.3. Collection of hemolymph and estimation of glucose

A 10 μL hemolymph sample was collected from the arthrodial membrane of coxa of third pair of walking leg using a micro-syringe and mixed with 250 μL of distilled water and then stored at −80 °C. Glucose concentration was measured enzymatically using the glucose oxidase assay kit (Sigma Chemical Co. St. Louis, MO, USA). The assay protocol essentially followed the one provided by the manufacturer. In brief, to the hemolymph sample, 0.5 mL of assay reagent was added and the contents were then incubated at 37 °C for 30 min. The reaction was arrested by the addition of 0.5 mL of 6.0 N sulphuric acid. After centrifuging at 2000*g* for 5 min the clear supernatant was spectrophotometrically quantified at a wavelength of 540 nm in UV–vis Spectrophotometer (Hitachi U-2001). Glucose concentrations were calculated on the basis of standard curve prepared using analar glucose solution.

### 2.4. Metal analysis

The hepatopancreas, chelate leg muscle and gill tissues were soaked on the filter paper and weighed immediately. The tissues (100 mg each) were digested over a sand bath twice with nitric acid (1.0 mL) and finally with 2.0 mL of acid mixture of nitric acid and perchloric acid (1:1). The digested samples were dissolved in 0.5 mL of 1% nitric acid and made up to 5 mL with glass distilled water. The hemolymph (0.5 mL) and medium was also digested in a similar fashion. The processed and digested samples were analyzed on Atomic Absorption Spectrophotometer (Shimadzu AA 6300) to determine the metal concentrations.

### 2.5. Statistical analysis of data

Data were expressed as mean ± SD. Statistical analysis was performed using an analysis of variance (one way ANOVA) followed by Tukey's post-test using SPSS (Student version 7.5, SPSS Inc, Chertsey, UK). Differences were considered to be significant at $p < 0.05$.

## 3. Results

The hemolymph glucose concentration in the metal-exposed intact crabs increased significantly ($p < 0.05$) compared to the hemolymph sugar level in the crabs in tap water (Table 1). The increase was more pronounced in the mercury-exposed crabs (117% above control) than in the cadmium-exposed ones (73% above control), a difference that was also statistically significant ($p < 0.05$) (Table 1).

Experiment II was performed to determine the role of eyestalks in the induction of hyperglycemia in crabs after exposure to metals. Bilateral eyestalk ablation (24 h) significantly decreased hemolymph glucose levels (Table 2). In contrast, the sugar concentration in the hemolymph of the eyestalkless crabs in metal-containing water did not change significantly when compared to eyestalkless crabs in metal-free water (Table 2).

Experiment III was performed to determine the effect of exposure to cadmium or mercury on hyperglycemic activity in the eyestalks. Injection of eyestalk extract of control crabs produced significant hyperglycemia (51% above control) (Table 3). Injection of either saline (25 μL) or extracts of eyestalks of cadmium- or mercury-exposed crabs did not cause any significant change in hemolymph glucose concentration when compared with uninjected crabs. The increases in the saline, eyestalk extract from cadmium- or mercury-exposed crabs were 0.83%, 18.48% or 11.29%,

**Table 1**
Effect of exposure to cadmium chloride or mercuric chloride on hemolymph glucose level of intact *O. senex senex*.

| Treatment | Hemolymph glucose level (mg glucose/100 mL) |
|---|---|
| Control | $16.43^{a} \pm 1.82$ |
| Cadmium | $28.37^{b} \pm 2.04$ (72.67) |
| Mercury | $35.66^{c} \pm 2.37$ (117.04) |

Values are mean ± S.D of 10 crabs.
Values in parentheses are % change from control.
Mean values that do not share the same superscript differ significantly at $p < 0.05$.

respectively, above control and the difference was not statistically significant ($p = 0.8639$; 0.064 and 0.0713, respectively) (Table 3).

The results of Experiment IV show the effects of 24 h exposure to cadmium or mercury on CHH-peptide concentration in the eyestalks. Eyestalks of mercury- or cadmium-exposed crabs contained significantly less CHH-immunoreactive peptide (46% and 52% below control) than the eyestalks of control crabs (Table 4).

### 3.1. Metal analyzes

Cadmium was present in appreciable quantity in the hepatopancreas, chelate leg muscle and gill tissues and hemolymph even in control animals and it is clear that after exposure to 0.24 ppm cadmium for 24 h resulted in significant amount of cadmium accumulation in these tissues (Table 5). The degree of accumulation of cadmium in the tissues of treated animals was observed as follows:

Gill > hepatopancreas > muscle.

Exposure to 0.07 ppm mercury for 24 h also resulted in significant accumulation of mercury in the tissues of crab and the degree of accumulation was observed as follows: gill > muscle > hepatopancreas (Table 5). The concentration of cadmium and mercury in the medium of exposure tanks was also determined before and 24 h after introducing the crabs. There was no significant change in both cadmium and mercury in the medium between the two time points (Table 5).

**Table 2**
Effect of eyestalk ablation (ESX) and exposure to cadmium chloride or mercuric chloride on hemolymph glucose level of ablated *O. senex senex*.

| Treatment | Hemolymph glucose level (mg glucose/100 mL) |
|---|---|
| Intact | $16.43^{a} \pm 1.82$ |
| Eyestalk ablated (ESX) | $8.74^{b} \pm 0.75$ ($-46.80$) |
| ESX-Cadmium | $8.97^{b} \pm 0.82$ (0.46) |
| ESX-Mercury | $8.21^{b} \pm 0.85$ (6.06) |

Values are mean ± S.D of 10 crabs.
Values in parentheses are % change from control.
Mean values that do not share the same superscript differ significantly at $p < 0.05$.
For calculation of % change and evaluation of '*p*' for ESX group intact served as control; for ESX-cadmium and ESX-mercury groups, ESX was used as control.

**Table 3**
Effect of metals on hemolymph glucose concentrations in crabs uninjected or injected with saline or various eyestalk extracts.

| Treatment | Hemolymph glucose level (mg glucose/100 mL) |
|---|---|
| Uninjected | $16.83^{a} \pm 1.78$ |
| Saline | $16.97^{a} \pm 1.82$ (0.83) |
| Control-ESE | $25.48^{b} \pm 2.01$ (51.39) |
| Cadmium-ESE | $19.94^{a} \pm 2.46$ (18.48) |
| Mercury-ESE | $18.73^{a} \pm 2.58$ (11.29) |

Values are mean ± S.D of 10 crabs.
Values in parentheses are % change from uninjected.
Mean values that do not share the same superscript differ significantly at $p < 0.05$.

**Table 4**
Effect of metals on levels of CHH peptides in eyestalks.

| Treatment | CHH peptide level (fmol/sinus gland) |
|---|---|
| Control | $67.09^{a} \pm 8.24$ |
| Cadmium | $32.15^{b} \pm 6.65$ ($-52.08$) |
| Mercury | $36.23^{b} \pm 7.87$ ($-45.99$) |

Values are mean ± S.D. $n = 5$ for each group.
Values in parentheses are % change from control.
Mean values that do not share the same superscript differ significantly at $p < 0.05$.

## 4. Discussion

In the crab, *O. senex senex*, eyestalk ablation results in hypoglycemia and injection of eyestalk extract induces significant hyperglycemia. It has also been reported earlier that *O. senex senex* eyestalk hyperglycemic hormone maintains glucose homeostasis (Kishori and Reddy, 2005). An elevation in glucose concentration in the hemolymph of *O. senex senex* following exposure to cadmium or mercury is apparently mediated by the eyestalk CHH hormone, since an increase in the hemolymph glucose concentration following metal exposure was absent after eyestalks were ablated. Both cadmium and mercury most likely rendered their hyperglycemic action through stimulating the release of CHH from the eyestalk neuroendocrine cells to the circulation. In support to this, the eyestalks from the crabs exposed to metals produced less hyperglycemia when compared to the eyestalks from the control crabs. The levels of CHH-immunoreactive peptide was also significantly less in the eyestalks of crabs exposed to metals when compared to the CHH levels in the eyestalks of control crabs.

Although metals-induced hyperglycemia is demonstrated in the freshwater crab *Barytelphusa cunicularis* (Machale et al., 1989), the fiddler crab *Uca pugilator* (Reddy et al., 1996a,b), the crayfish *P. clarkii* (Reddy et al., 1994) and the freshwater prawn *Macrobrachium kistnensis* (Nagabhushanam and Kulkarni, 1981), the present study demonstrates that they are involved in altering the levels of hyperglycemic hormone. Release of CHH from the eyestalks of American lobster *Homarus americanus* was also observed following emersion, exposure to high temperatures and salinity stress (Chang et al., 1998). Webster (1996) found that emersion causes a significant increase in CHH in the hemolymph of edible crab, *Cancer pagurus*.

Earlier, Clare et al. (1992) and Weis et al. (1992) observed inhibitory effects of metals on molting and regeneration of limbs, which are neuroendocrine-mediated processes, and suggested the use of these processes as biomarkers for developmental, physiological and endocrinological effects of pollutants on crustaceans. An increase in CHH secretion by metals, as the current results

**Table 5**
Concentration of cadmium and mercury in medium and in different tissues before and 24 h after exposure to metals.

| Group | Medium (μg/100 mL) | | Hepatopancreas (μg/g) | | Muscle (μg/g) | | Gill (μg/g) | | Hemolymph (μg/100 mL) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 0 h | 24 h | 0 h | 24 h | 0 h | 24 h | 0 h | 24 h | 0 h | 24 h |
| | | | — | — | — | — | — | — | — | — |
| Cadmium | $24.48^{a} \pm 2.09$ | $24.01^{a} \pm 2.16$ | $0.14^{a} \pm 0.04$ | $0.37^{b} \pm 0.05$ | $0.06^{a} \pm 0.01$ | $0.16^{b} \pm 0.03$ | $0.16^{a} \pm 0.04$ | $0.52^{b} \pm 0.07$ | $0.14^{a} \pm 0.05$ | $0.55^{b} \pm 0.08$ |
| Mercury | $7.38^{a} \pm 1.26$ | $7.11^{a} \pm 1.18$ | $0.08^{a} \pm 0.01$ | $0.12^{b} \pm 0.02$ | — | $0.06 \pm 0.01$ | $0.02^{a} \pm 0.01$ | $0.19^{b} \pm 0.04$ | — | $0.21 \pm 0.06$ |

Values are mean ± S.D. $n = 8$ for each group.
Mean values that do not share the same superscript differ significantly at $p < 0.05$.

suggest, is consistent with the results reported by Reddy et al., (1996a,b) for the *Uca pugilator* where cadmium-produced hyperglycemia presumably by triggering secretion of the CHH. Also, cadmium inhibits molting (Weis, 1976) and reproduction in crabs (Reddy et al., 1997; Rodriguez et al., 2000) quite likely by increasing molt-inhibiting (MIH) and gonad-inhibiting hormone (GIH) secretion from eyestalks. CHH, MIH and GIH belong to the same family of peptide neurohormones in the eyestalk (Chang, 1997) synthesized in the X-organ sinus gland complex, stored in and released from sinus gland. Perhaps a general action of metal is to cause hypersecretion of the members of the CHH family neuropeptides.

It seems evident, therefore, that exposure to metals induces hyperglycemia in intact crabs, *O. senex senex*. However, metal had no significant effect on the hemolymph glucose level in eyestalkless crabs. The eyestalks of metal-exposed crabs produced less hyperglycemia and possessed lower concentrations of CHH peptide when compared to the eyestalks from control crabs. These results strongly suggest that hyperglycemia caused by metals in intact crabs was due to the triggering release of CHH in the crab.

The study also demonstrates the uptake of cadmium and mercury in different tissues of crab after exposure to sublethal concentrations of metals. Though significant accumulation of metals was observed in the tissues of crab after 24 h, the metal concentration was not significantly altered in the medium. The non-significant changes in the metal concentration observed in the medium even after introducing animals could be due to the volume of medium per crab used in the present study. The volume of medium per crab with a body weight of 30 g was maintained at 10 L. This might be the reason for the negligible changes in the metal concentration in the medium though there was a significant accumulation of metals in the tissues.

## 5. Conclusion

On the basis of the results obtained in this study it can be concluded that exposure to sublethal concentrations of cadmium or mercury resulted in significant hyperglycemia in intact crabs, but not in eyestalk ablated crabs suggesting that metal-induced hyperglycemia is mediated through eyestalk hyperglycemic hormone. The levels of hyperglycemic hormone and the hyperglycemic effect was significantly lower in the eyestalks collected from metal-exposed crabs when compared with eyestalks from control crabs, strongly suggesting that metals act, at least in part, by triggering secretion of hyperglycemic hormone from the eyestalk. In conclusion, we have demonstrated for the first time that the eyestalk neuroendocrine system is involved in metal-induced hyperglycemia in the crab *O. senex senex*.

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## Animal procedures

All animal procedures were approved by the Institutional Review Board and Institutional Animal Care and Ethics Committee at S.V. University, Tirupati. The authors declare that the experiments done during these studies comply with the laws of their Country.

## Declaration

The authors declare that there is no conflict of interest that would prejudice the impartiality of this scientific work.

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